House Price Prediction – Summary

Objective

The objective of this project was to predict house prices using machine learning techniques based on property features such as area, bedrooms, bathrooms, stories, parking spaces, and housing amenities.

Data Preparation

The dataset was cleaned by checking for missing values, removing duplicate records, and converting categorical variables into numerical format using one-hot encoding. The data was then split into training and testing sets using an 80:20 ratio.

Model Performance

1. **Linear Regression**
    - MAE: 970,043
    - RMSE: 1,324,507
    - R^2 Score: 0.653
2. **Random Forest Regressor**
    - MAE: 1,013,365
    - RMSE: 1,389,776
    - R^2 Score: 0.618

Key Findings

Area, bathrooms, stories, and housing amenities had the strongest influence on house prices. The price distribution was right-skewed, indicating that most houses belonged to the lower and middle price ranges. A notable finding was that Linear Regression outperformed Random Forest on this dataset.

Conclusion

Linear Regression was selected as the final model because it achieved the highest predictive performance. The project successfully demonstrated data preprocessing, visualization, machine learning model development, and evaluation for house price prediction.